

Venue Activity Prediction: a model for Social Media Integration (working title)

Your name

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Abstract

Placeholder. Write the abstract last, once the model and the simulation results exist.

1 Introduction

2 Literature Review

2.1 Search methodology

Claude searched the literature against three research questions, stated here in full since the trimming below depends on them:

- RQ1. How do you estimate the true size or activity of a population at a location when only a fraction is observed?
- RQ2. How do you infer venue occupancy from event data when the observed users are a non-uniform sample?
- RQ3. Which point-process and arrival models describe bursty, clustered arrivals, and how do they treat groups rather than independent individuals?

This search returned approximately 90 candidate references.

2.2 Trimming the candidate set

Reading 90 papers in full is not feasible at this stage. The set was trimmed to the 8 references that the search itself flagged as shaping the method most directly, leaving the rest at abstract-level triage, to revisit only if a specific gap in the model requires it. Each entry below is still a lead, not a checked source; every one needs to be opened and logged in `AI/SOURCES.md` before any claim in this paper cites it.

- Brown et al. (2014), on defining a group check-in by a time window. No confirmed bibliographic detail (DOI, venue, page range) has surfaced yet; the paper still needs locating, not just reading.
- Chorley et al. (2016), *ICWSM*, the main prior study of Untappd check-in data. Full bibliographic detail is likewise not yet confirmed.
- Cho, Ver Steeg, and Galstyan (2014), *AAAI* 28(1):269–275, on a per-venue Hawkes model separating self-excitation, social, and exogenous effects in check-in data.
- Clement, Converse, and Royle (2017), *Ecological and Evolution* 7(18):7304–7310, on two-stage detection: whether a group is observed at all, and how many of its members are.

- Zhuang and Mateu (2019), *JRSS A* 182(3):919–942, on separating a periodic seasonal background from genuine self-excitation in clustered event data.
- Yauck, Rivest, and Rothman (2019), *JASA* 114(525):105–114, on capture-recapture applied to app-activation traces to estimate unique visitors to a commercial venue.
- Timokhin, Sadrani, and Antoniou (2020), *Smart Cities* 3(3):818–841, on validating a popularity signal against an independent ground truth, a low-cost recipe for exactly the ground-truth problem this project’s synthetic simulator is meant to solve by construction instead.
- Daw and Pender (2019), *Queueing Systems* 91(3–4):367–401, on how batch arrivals inflate occupancy variance relative to individual arrivals with the same mean.

3 Problem Statement

4 Model

5 Simulation

6 Results

7 Discussion

References